NOTE: MIL-STD-2191 has been redesignated as a Standard Practice. The cover page has been changed for Administrative reasons. There are no other changes to this Document.

INCH-POUND

MIL-STD-2191 (SH)
5 NOVEMBER 1992
SUPERSEDING
NAVSEA 0900-LP-014-1010
JUNE 1966

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

REPAIR WELDING, WELD CLADDING, STRAIGHTENING, AND COLD ROLLING OR MAIN PROPULSION SHAFTING

AMSC N/A FSC 2010

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FOREWORD

1. This military standard is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard contains minimum requirements for weld cladding, repair welding, mechanical straightening, and cold rolling of new and refurbished main propulsion shafting and for repair welding of rudder stocks for ships of the U.S. Navy.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 14.2).

SPECIFICATIONS

MILITARY

MIL-S-13165	- Shot Peening of Metal Parts.
MIL-E-21562	- Electrodes and Rods - Welding, Bare, Nickel Alloy.
MIL-E-22200/1	- Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, as Welded or Stress-Relieved Weld Application.
MIL-E-22200/3	- Electrodes, Welding, Covered: Nickel Base Alloy; and Cobalt Base Alloy.
MIL-E-22200/10	- Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium, High Tensile and Higher-Strength Low Alloy Steels.
MIL-S-23284	- Steel Forgings, Carbon and Alloy, for Shafts, Sleeves, Propeller Nuts, Couplings, and Stocks (Rudders and Diving Planes).
MIL-E-23765/1	- Electrodes and Rods - Welding, Bare, Solid and Alloyed Cored, Ordinary Strength and Low Alloy Steel.
MIL-E-23765/2	- Electrodes and Rods - Welding, Bare, Solid, or Alloyed Cored, Low Alloy Steel.
MIL-E-24403/1	- Electrodes - Welding, Flux Cored, Ordinary Strength and Low Alloy Steel.
MIL-E-24403/2	- Electrodes - Welding, Flux Cored, Low-Alloy Steel.
MIL-R-81841	- Rotary Flap Peening of Metal Parts.

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STANDARDS

MILITARY

MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification.
MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-278 - Welding and Casting Standard.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-003-8000 - Metals, Surface Inspection Acceptance Standards.
0900-LP-006-3010 - Hull Structure Production - Repair Welds Ultrasonic Inspection Procedure - Acceptance Standards for.
0910-LP-105-1300 - SSN and SSB(N) Submarine Main Propulsion Shafting, TRS Volume Part 1 Refurbishment Procedures. (S9243-AT-TRS-011/0203-086-009A).
0910-LP-105-1400 - Surface Ships Main Propulsion Shafting, TRS Volume 1 Part 1 Refurbishment Procedures. (S9243-AW-TRS-011/0203-086-501A).

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 14.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 8 - Standard Test Methods of Tension Testing of Metallic Materials. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A3.0 - Standard Welding Terms and Definitions Including Terms for Brazing, Soldering Thermal Spraying and Thermal Cutting. (DoD adopted)
A5.01 - Filler Metal Procurement Guidelines.

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(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 General. Except as noted herein, welding nomenclature and definitions shall conform to AWS A3.0.

3.1.1 Weld build-up (build-up). Weld build-up is depositing weld metal that is compatible to the base material in metallurgical and mechanical properties for the purpose of restoring welded dimensions, thicknesses, or other configuration requirements.

3.1.2 Cladding. Cladding is a surfacing variation that deposits or applies surfacing material, usually to improve corrosion or heat resistance.

3.1.3 Completed weld. Welding is completed when preheat is removed and the material has cooled to ambient temperature and the weld has been visually accepted and is ready for other nondestructive test (NDT) inspections.

4. GENERAL REQUIREMENTS

4.1 Repairs. Repairs shall include replacement of shafting sections or flanges, building up of eroded or corroded areas, repair of damaged bolt holes or keyways, and repair of cracks. Repairs shall also include rework by grinding.

4.1.1 Major repairs. Repair shall be considered major if it involves joining two sections of shafts together, replacing a flange, repair of cracks 1/2 inch deep or deeper, build-up of 3/8 inch or more in thickness, or build-up of any thickness that covers 5 percent or more of the total surface area in way of sleeves, tapers, flanges or fillets, or build-up of any thickness that cover more than 25 percent of the remaining surface area. Grooves from cable wrapping are to be considered cracks for classification as to major or minor repairs. Thicknesses are defined as final machined thickness.

4.1.1.1 Welding of major repairs. Unless otherwise approved (see 14.4), welding of repairs classified as major shall be accomplished after extending the weld repair groove 360 degrees around the shaft circumference to minimize distortion and equalize weld stresses.

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4.1.2 Minor repairs. Repairs shall be considered minor if they involve build up of keyways and bolt holes or otherwise do not meet the criteria specified in 4.2.1.

4.1.3 Cracks. Crack repairs, whether classified minor or major, or repaired by fairing or welding, shall be reported (see 14.4).

4.1.4 Cold rolling after welding or straightening. Cold rolling may be required after welding or straightening (see 12.1.1 and 12.1.2).

4.2 Shaft material. This standard shall be followed for repair of all steel shafts. Repairs to shafts of materials not listed in 4.2.1 or 4.2.2 require approval (see 14.4).

4.2.1 Carbon steels. Replacement (new) shafts of carbon steels shall be in accordance with MIL-S-23284, class 3 and 4. For the purpose of welding qualification, class 3 and class 4 will be considered S-1 in accordance with MIL-STD-278.

4.2.2 Alloy steels. Replacement (new) shafts of alloy steels shall be in accordance with MIL-S-23284, class 1, 2, 5, and 6.

4.2.3 Other shaft materials. When shaft material cannot be determined from markings on the shaft, or for materials other than as specified in 4.3.1 or 4.3.2, the following information shall be determined prior to welding, mechanical straightening, or cold rolling (see 14.4).

(a) Hardness information including a minimum of five hardness readings taken from the end of the shaft. Standard Rockwell (B or C scale) or Brinell (standard ball) hardness testing is acceptable. Microhardness (for example, Diamond pyramid) testing is not acceptable.
(b) Chemical composition information from an unwelded portion of the shaft. The minimum elements analyzed shall include: carbon (C), manganese (Mn), silicon (Si), chromium (Cr), molybdenum (Mo), nickel (Ni), vanadium (V), sulfur (S), and phosphorus (P). The weight percent shall be reported to two decimal places (X.XX) for all elements except P and S, which shall be reported to three decimal places (X.XXX).
(c) Known information listed in 10.1.1.

5. WELDING REQUIREMENTS

5.1 Welding procedure qualification. Procedures for welding and cladding shall be qualified in accordance with MIL-STD-248 and as specified in 5.1.1 through 5.1.1.5.2. Procedures and personnel previously qualified to earlier revisions of this standard, for which records are available and qualifications have been maintained, shall not in general require requalification to this revision. However, the authorized representative may require requalification to this revision on a case-by-case basis where he has specific concerns about the adequacy of procedures or ability of personnel to produce welds of intended integrity or which meet requirements of the applicable fabrication document.

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5.1.1 Specific qualification requirements.

5.1.1.1 Essential elements. Unless otherwise approved (see 14.4), for weld depth greater than 2 inches the joint designs on figure 1 and figure 2 shall be utilized. These figures may also be used for weld depth of 2 inches or less. For all repair welds, weld joint geometry shall be selected to minimize the volume of weld metal deposited. In addition to qualification requirements in accordance with MIL-STD-248, the following shall be included as essential elements of the weld procedure.

(a) Welding sequence (selected to minimize distortion).
(b) Rotation (circumferential) speed of the shaft for the horizontal rolled position for automatic, mechanized or semiautomatic welding processes.
(c) Longitudinal (axial) speed of the head for automatic or mechanized welding processes.
(d) Distortion control methods (for example, peening).
(e) Methods of final inspection.
(f) Method of stress relief (see 10.4.5).
(g) Rotation speed for horizontal rolled stress relief (see 10.4.5.1).
(h) Method for preventing surface scaling during stress relief (see 10.4.5.2).
(i) Method of measuring distortion during welding (see 10.4.8).

Note: Change in rotational and longitudinal speed will not require requalification as long as the heat input does not exceed heat input used for qualification.

5.1.1.2 Inspection and test requirements. Destructive and nondestructive testing shall be performed after stress relief but no sooner than 24 hours after the material has reached ambient temperature.

5.1.1.2.1 Welding qualifications. Weld procedure qualification shall be in accordance with MIL-STD-248, with the following exceptions:

(a) Radiographic inspection (RT) is not required.
(b) Ultrasonic inspection (UT) as specified in 7.3 is required.
(c) Two transverse weld and two all weld metal tensile specimens are required.
(d) Two macrographic specimens, removed transverse to the direction of welding, meeting requirements in accordance with MIL-STD-248 are required.
(e) A microhardness survey covering weld metal, heat affected zone (HAZ), and base metal from both sides of the weld deposit is required. The survey shall traverse perpendicular to the centerline of the weld. Linear distance between hardness testing sites shall be not greater than 0.020 inch. The survey shall indicate a relatively uniform hardness level across the area tested. Final evaluation of the hardness profile shall be sought (see 14.4).
(f) Sets of five impact specimens shall be prepared for the weld metal, sets of three impact specimens shall be prepared from the heat affected zone (HAZ) of each base metal and sets of three impact specimens shall be prepared from the non-affected area of each

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base metal. Impact testing shall be performed to the requirements of MIL-STD-248. Impact specimens shall meet impact requirements of the base metal at the temperature listed in the base metal material specification. If the base material specification does not require impact testing, approval shall be sought (see 14.4).

5.1.1.2.2 Cladding qualifications. Cladding procedure qualification shall be in accordance with MIL-STD-248, with the following exceptions:

(a) Ultrasonic inspection (UT) as specified in 7.3 is required.
(b) When the procedure involves corrosion-resistant cladding, chemical analysis of the clad material at the minimum thickness qualified of cladding is required. For Inconel 625, iron content shall be not greater than 9 percent and molybdenum content shall be not less than 8 percent.
(c) Three tensile tests of the clad material are required. Test specimens shall be sub-size, flat, 1/8 inch thick in accordance with ASTM E 8, and shall be taken after final heat treatment, normal to the direction of welding, at the minimum thickness qualified for cladding. Minimum acceptable yield strength shall be 60 kilopounds per square inch (klb/in^2) at 0.2 percent offset. Other mechanical properties shall be as required by the electrode specification.

5.1.1.3 Test assembly. The test assembly shall be in accordance with MIL-STD-248, with the following additional requirements:

(a) The test assembly shall be cylindrical.
(b) The test assembly shall be a shaft when major repairs are required in other than cladding or buildup.
(c) As approved (see 14.4) for noncircumferential minor repairs.

5.1.1.4 Thickness. A 1-1/2 inch weld groove in a 3-inch (minimum) wall thickness hollow cylinder or a 1-1/2 inch weld groove in a 6-inch (minimum) diameter solid cylinder shall qualify for all thicknesses of material.

5.1.1.5 Cross qualification.

5.1.1.5.1 MIL-S-23284 materials. Welding procedure cross qualification for MIL-S-23284 materials shall be allowed as follows:

(a) Welding of class 1 material shall qualify for welding of class 1 and class 2 material.
(b) Welding of class 2 material shall qualify for welding of only class 2 material.
(c) Welding of class 3 material shall qualify for welding of class 3 and class 4 material.
(d) Welding of class 4 material shall qualify for welding of only class 4 material.
(e) Welding of class 6 material shall qualify for welding of class 6 and class 5 material.
(f) Welding of class 5 material shall qualify for welding of only class 5 material.

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5.1.1.5.2 Materials not listed in MIL-S-23284. Welding procedure cross qualification for materials other than as listed in 4.2.1 and 4.2.2 shall not be allowed except as specifically approved on a case basis (see 14.4). For cross qualification approval on a case basis, all welding requirements listed herein shall be met. In addition, approval shall be sought (see 14.4) for the following information prior to weld repair:

(a) For the shaft to be repaired, information listed in 4.2.3.
(b) The applicable welding procedure prepared as specified in 5.1.1.1.
(c) The applicable welding procedure qualification information prepared as specified in 5.1.1.2, 5.1.1.3, and 5.1.1.4.

5.2 Welder performance qualification. Prior to performing any production welding covered by this standard, personnel shall be qualified in accordance with MIL-STD-248. Changes or modifications to distortion control aspects of welding procedures do not require requalification of welders; however, they shall be thoroughly instructed in such changes prior to production use of the modified procedure.

6. RECORD REQUIREMENTS

6.1 Records. The quality control system shall include the preparation and maintenance of written records. Records shall contain, as a minimum, the following information. (Vendor inspection or mill certificate records will fulfill the requirements of this section for that portion of the required information contained therein.)

(a) Welding procedure qualification records in accordance with MIL-STD-248 and as required herein.
(b) Welder and welding operator qualification records in accordance with MIL-STD-248.
(c) Nondestructive test (NDT) procedure qualification records in accordance with MIL-STD-271.
(d) NDT personnel qualification records in accordance with MIL-STD-271.

Unless otherwise specified, all required records shall be maintained by the activity and be available to NAVSEA or its authorized representative throughout the life of the contract and for 3 years after delivery. At the expiration of the record retention period, NAVSEA or its authorized representatives shall be given a written notification. Disposition of records shall be as agreed upon by NAVSEA and the contractor.

6.1.1 Welding electrode control inspection. Records of inspection required by section 8 shall contain the following:

(a) Manufacturer's certification and test results for compliance with the applicable specification.
(b) Date of inspection and signature of inspector.

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6.1.2 Repairs. Records of repair inspection shall contain the following:

(a) Identification of shaft (serial number, ship).
(b) Shaft material (chemical analysis as specified in 4.2.3 for materials not listed in 4.2.1 or 4.2.2).
(c) Location, size, and orientation of defects.
(d) Date and type of original inspection and inspector's signature.
(e) Welding procedure and filler material used.
(f) Welder's identification and signature.
(g) Post weld thermal treatment including charts or recordings of times and temperatures.
(h) In process and final inspections, dates, results, and inspector's identification and signature.

6.1.3 Nondestructive testing. In addition to nondestructive testing record requirements in accordance with MIL-STD-271, records of nondestructive testing shall include the following:

(a) Component description and unique identification.
(b) Approved procedure identification.
(c) Equipment/material used.
(d) Acceptance standard used.
(e) Date of inspection.
(f) Inspector's signature.

7. INSPECTION REQUIREMENTS

7.1 Visual inspection. Visual inspection shall be performed in accordance with MIL-STD-271. Inspection shall be performed as follows:

(a) Areas faired in or prepared for weld repair shall be inspected to ensure defects have been removed and the area blends smoothly into the base material.
(b) Each layer of weld deposit shall be inspected. Cracks and lack-of-fusion (including weld rollover) shall be rejected. The final weld surface shall meet class I weld acceptance criteria of NAVSEA 0900-LP-003-8000.

7.2 Magnetic particle and liquid penetrant inspection. Ferromagnetic material shall be magnetic particle inspected. Nonferromagnetic material, or applications where magnetic particle inspection is not practical, shall be liquid penetrant inspected. Inspections shall be performed in accordance with MIL-STD-271. The following areas shall be inspected:

(a) Defect removal sites.
(b) Completed weld repairs and weld cladding (inspect no sooner than 24 hours after final heat treatment).
(c) Machined areas (including areas repaired by fairing in).
(d) Areas subject to temperatures above 500 degrees Fahrenheit (°F) (for example hot-spot straightening).
(e) Areas mechanically worked (for example, peening, rolling).

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Surfaces shall meet requirements of NAVSEA 0900-LP-003-8000. For welds, class I criteria shall apply. Clad areas shall be evaluated to the weld criteria, not the weld overlay requirements. The shaft shall be demagnetized after magnetic particle testing.

7.3 Ultrasonic inspection. Inspection shall be performed as follows:

(a) Clad surfaces and weld build-up areas comprised of more than one weld layer (more than one pass) shall be inspected using the ultrasonic inspection procedure for weld deposited buttering in accordance with MIL-STD-278. If the weld metal and base material are acoustically similar (as defined in MIL-STD-271), it is not necessary to apply weld metal on the calibration block. Inspection shall be performed after post-weld heat treatment. Acceptance criteria shall be in accordance with MIL-STD-278 for weld deposited buttering using the weld thickness as the design material thickness.

(b) Other weld repairs (for example, weld repair of cracks) shall be inspected to the ultrasonic weld inspection requirements in accordance with MIL-STD-271. Inspection shall not be performed sooner than 24 hours after post-weld heat treatment. Acceptance criteria shall be in accordance with class I of NAVSEA 0900-LP-006-3010. Criteria shall be based on the weld metal thickness or 1 inch, whichever is less. For indications within 1/4 inch of any surface, any indication whose amplitude equals or exceeds the disregard level (DRL) shall be rejected.

8. RECEIPT INSPECTION OF WELDING FILLER MATERIALS

8.1 Responsibility. The receiving activity shall be responsible for verifying that the inspections required in this section are performed and documented. Certified test results for all filler materials, showing chemical and mechanical properties, shall be obtained and reviewed for compliance with the applicable material specification. Level I filler materials received from Navy Ships Parts Control Center (SPCC) by Naval shipyards shall be considered as meeting the chemical and mechanical properties of the filler material, as documented by SPCC.

8.1.1 Contractor inspection. Where the required inspection is performed by the contractor or by other subcontracted representative prior to receipt, the receipt inspection by the receiving activity shall consist of checking the required documentation against the material received for compliance with the applicable material specification.

8.1.2 Receiving activity inspection. Where the required inspection has not been made prior to receipt, the receiving activity shall perform and record the inspections required by the applicable specification.

8.2 Nonconforming filler materials. Filler materials which do not comply with the applicable specification shall be isolated and not used for production welding.

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9. WELDING MATERIALS AND PROCESSES

9.1 Approved filler materials and processes. Representative filler material types and specifications shall be as specified in table I. The use of other processes and materials will be permitted, with approval (see 14.4), for the intended application. Unless otherwise specifically approved (see 14.4), the maximum heat input shall be 55,000 joules per inch for materials listed in 4.2.2.

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TABLE I. Filler materials and specifications. 1/

	MIL-S-23284 Class 1	MIL-S-23284 Class 2	MIL-S-23284 Class 3	MIL-S-23284 Class 4	MIL-S-23284 Class 5	MIL-S-23284 Class 6	Corrosion resistant cladding 6/
Shielded metal arc welding (SMAW)	MIL-E-22200/10 MIL-10018-M1	MIL-E-22200/10 MIL-10018-M1	MIL-E-22200/10 MIL-10018-M1	MIL-E-22200/10 MIL-7018-M	MIL-E-22200/10 MIL-10018-M1	MIL-E-22200/10 MIL-12018-M2	MIL-E-22200/3 MIL-1N12
Gas metal arc welding (GMAW)	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-120S-2	MIL-E-21562 MIL-EN625
Gas tungsten arc welding (GTAW)	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-120S-2	MIL-E-21562 MIL-EN625
Flux cored arc welding (FCAW)	MIL-E-24403/2 MIL-100TC MIL-100TM MIL-100TS	MIL-E-24403/2 MIL-100TC MIL-100TM MIL-100TS	MIL-E-24403/1 MIL-80T1-Ni1 MIL-80T1-Ni2	MIL-E-24403/1 MIL-70T-5 MIL-70T-8	MIL-E-24403/2 MIL-100TC MIL-100TM MIL-100TS	MIL-E-24403/2 MIL-110TC	
Submerged arc welding (SAW)	MIL-E-23765/2 MIL-100S-2 MIL-100S-2F	MIL-E-23765/2 MIL-100S-2 MIL-100S-2F	MIL-E-23765/2 MIL-100S-2 MIL-100S-2F	MIL-E-23765/1 MIL-70S-2 MIL-70S-2F	MIL-E-23765/2 MIL-100S-2 MIL-100S-2F	MIL-E-23765/2 MIL-120S-2 MIL-120S-2F	MIL-E-21562 MIL-EN625 Neutral flux

See footnotes at end of table.

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TABLE I. Filler materials and specifications - Continued.

STEELS NOT INCLUDED IN MIL-S-23284 OR OBLITERATED SHAFT MARKINGS 2/

Maximum tensile strength Maximum yield strength	Greater than 95,000 lb/in² Greater than 65,000 lb/in² 1/	Maximum tensile 95,000 lb/in² Maximum yield 65,000 lb/in²	Maximum tensile 80,000 lb/in² Maximum yield 45,000 lb/in²	Maximum tensile 75,000 lb/in² Maximum yield 40,000 lb/in²	Maximum tensile 60,000 lb/in² Maximum yield 30,000 lb/in²	Maximum tensile 50,000 lb/in² ----------	Corrosion resistant cladding 6/
Shielded metal arc welding (SMAW)	3/	MIL-E-22200/1 MIL-10018-M 4/	MIL-E-22200/1 MIL-8018-C3	MIL-E-22200/1 MIL-8018-C3	MIL-E-22200/1 MIL-7018 5/	MIL-E-22200/1 MIL-7018 5/	MIL-E-22200/3 MIL-1N12
Gas metal arc welding (GMAW)	3/	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-80S-2	MIL-E-23765/2 MIL-80S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-21562 MIL-EN625
Gas tungsten arc welding (GTAW)	3/	MIL-E-23765/2 MIL-100S-2	MIL-E-23765/2 MIL-80S-2	MIL-E-23765/2 MIL-80S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-21562 MIL-EN625
Flux cored arc welding (FCAW)	3/	MIL-E-24403/2 MIL-100TC MIL-100TM MIL-100TS	MIL-E-24403/1 MIL-80T1-N11 MIL-80T1-N12	MIL-E-24403/1 MIL-70T-5 MIL-70T-8	MIL-E-24403/1 MIL-70T-6 MIL-70T-1	MIL-E-24403/1 MIL-70T-6 MIL-70T-1	
Submerged arc welding (SAW)	3/	MIL-E-23765/2 MIL-100S-2 MIL-100S-2F	MIL-E-23765/2 MIL-80S-2 MIL-80S-2F	MIL-E-23765/2 MIL-80S-2 MIL-80S-2F	MIL-E-23765/1 MIL-70S-2 MIL-70S-2F	MIL-E-23765/1 MIL-70S-2 MIL-70S-2F	MIL-E-21562 MIL-EN625 Neutral flux

See footnotes at end of table.

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1/ Equivalent filler materials meeting AWS specifications may be substituted for MIL-type filler materials. Acquisition of AWS electrodes shall be in accordance with AWS A5.01, lot classification C4 and S3, for covered and bare electrodes, respectively, and a certified itemized test report in accordance with schedule J is required.
2/ See 4.2 and 4.2.3.
3/ Filler materials selected for these steels shall be as approved on a case basis (see 14.4).
4/ MIL-10018-M1 conforming to MIL-E-22200/10 is an acceptable alternate.
5/ MIL-7018-M conforming to MIL-E-22200/10 is an acceptable alternate.
6/ Corrosion resistant cladding consumables are applicable for all shaft materials for steels included or not included in MIL-S-23284 or shafts with obliterated shaft markings.
7/ Pounds per square inch (lb/in^2).

10. REPAIR REQUIREMENTS

10.1 Approval.

10.1.1 Major repairs. Approval is required for major repairs (see 14.4) and requires the following information:

(a) Military material specification number, revision number, and alloy class, as in MIL-S-23284-RX-CLX.
(b) Heat or forging number (traceability code).
(c) Shaft manufacturer's name, trademark, or symbol.
(d) Final tempering temperature.
(e) Stress relief temperature.
(f) Drawing or die number.
(g) Extent of repair required.
(h) Description of defects (type, size, and location).
(i) Welding repair procedure and qualification data.

10.1.2 Minor repairs. Approval for minor repairs shall be obtained from the authorized representative and requires all information listed in 10.1.1.

10.2 Defect removal. Defective areas shall be removed completely by chipping, grinding, or machining to sound base metal. Grooves or repair areas caused by removal of defects, and surfaces of the shaft adjacent thereto, shall be inspected as specified in section 7.

10.3 Repair by fairing in. General requirements for repair by fairing in shall be as specified in 10.3.1 through 10.3.5.

10.3.1 Minor pitting. Minor pitting, 1/16 inch in depth or less, shall be removed by using a rotary file, wire brush, or similar hand tool. In no case shall the resulting shaft diameter be more than 1/8 inch less than the minimum diameter shown on the applicable drawing.

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10.3.2 Cracks and severe pits. Cracks or severe pits 1/8 inch in depth or less for shafts up to and including 10 inches in diameter, or not exceeding 1/4 inch in depth or less for shafts exceeding 10 inches in diameter, may be repaired by grinding and dressing the ground areas to a large faired radius (three times the depth) to avoid stress concentrations. In no case shall the resulting shaft diameter be less than the minimum diameter shown on the applicable drawing by more than 1/4 inch for shafts 10 inches and under in diameter or 1/2 inch for shafts over 10 inches.

10.3.3 Additional crack and pitting requirements. The requirements specified in 10.3.1 and 10.3.2 shall apply to random cracks or pitting and shall be interpreted as allowing grooving to a continuous length of 180 degrees maximum along the shaft circumference. Grooving is not permitted where grooves are 180 degrees diametrically opposite each other. In addition, in any 3 foot length of shaft no combination of grooves shall addictively combine to exceed a length of 180 degrees along the shaft circumference.

10.3.4 Submarine shafts. For submarine shafts, additional requirements and restrictions on repair by fairing in shall be in accordance with NAVSEA 0910-LP-105-1300 (S9243-AT-TRS-011/0203-086-009A).

10.3.5 Surface ship shafts. For surface ship shafts, additional requirements and restrictions on repair by fairing in shall be in accordance with NAVSEA 0910-LP-105-1400 (S9243-AW-TRS/0203-086-501A).

10.4 Weld repair.

10.4.1 Materials requiring approval. Materials not listed in 4.2.1 or 4.2.2, and materials not identifiable by markings on the shaft, shall not be weld repaired without prior approval (see 4.2.3 and 14.4).

10.4.2 Preparation for welding. Areas to be weld repaired and adjacent base material within 1 inch of the weld repair area shall be free of oil, grease, or other surface contaminants which could adversely affect the welding process. Preparation shall provide adequate accessibility for welding.

10.4.3 Preheat. Weld joints and areas shall be preheated to the temperatures specified in table II. Preheat shall be applied in a manner that ensures uniform heating of the weld area and a minimum of 6 inches of base material on all sides of the repair areas. Preheat temperature shall be maintained throughout the welding operation.

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TABLE II. Preheating and post-weld heat treatment requirements.

Shaft material		Preheat-interpass temperature (°F)	Post-weld treatment temperature (°F) 1/ 2/ 3/ 4/
Specification	Class		
MIL-S-23284	1	400 min - 500 max	1200 ± 25
	2	400 min - 500 max	1200 ± 25
	3	400 min - 500 max	1175 ± 25
	4	275 min - 400 max	1175 ± 25
	5	275 min - 400 max	1050 ± 25
	6	275 min - 400 max	1050 ± 25

1/ Holding time at stress-relief temperature shall be 2 hours per inch of weld deposit thickness. A minimum holding time of 2 hours is required for welds less than 1 inch thick.
2/ Heating and cooling rates of shafting, as determined by thermocouples attached thereto shall be 150°F per hour maximum. The shaft may be cooled in still air after reaching 450°F (see 10.4.5.3).
3/ Maximum furnace temperature for charging shaft shall be 500°F.
4/ Post-weld heat treatment temperature must be a minimum of 50°F below tempering temperature. When post-weld heat treatment temperature is above actual shaft tempering temperature, post-weld heat treatment shall require specific approval (see 14.4). Activities shall determine tempering temperature of all shafts prior to post-weld heat treatment. Where tempering temperature information is not available, hardness surveys as specified in 10.4.6 shall be performed before and after stress relief.

10.4.4 Welding sequence. The welding sequence shall minimize distortion.

10.4.5 Post-weld heat treatment. After welding, shafts shall be stress-relief heat treated as specified in table II.

10.4.5.1 Stress relief position. Where practical, shafts shall be stress relief heat treated in the vertical position or in the horizontal position with the shaft rotated continuously while the shaft is above 500°F to minimize distortion. Repaired shafts shall meet straightness tolerances specified in the original or applicable shaft drawing following final heat treatment, unless otherwise approved (see 14.4).

10.4.5.2 Heating methods. Where practical, post-weld stress relief shall be accomplished by furnace heating. Local heating with resistance or induction type portable heating units is acceptable, provided the width of the heated area on each side of the weld groove is at least twice the weld thickness. Other heating methods require approval (see 14.4). Sections of shafts, including bores, which are anticipated to be above 500°F during stress relief, shall be covered with a

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heat resisting scale preventative such as "No Carb" or "No Scale" or equal. If coating of the bore is not practical, the bore may be plugged on each end and purged through the plugs with an inert gas. Care must be taken to allow for expansion of gases in the shaft bore during stress relief.

10.4.5.3 Temperature measurement during stress relief. Regardless of stress relief method, stress relief temperature shall be monitored by thermocouples directly attached to the weld repair area. A minimum of three thermocouples shall be required for each weld repair area. One thermocouple shall be located on the surface of the shaft on the centerline of the weld. The other thermocouples shall be located on either side of the weld repair, 3 inches from the weld. Thermocouples shall be separated from each other by a maximum of 120 degrees along the circumference of the shaft. Autographic reports (strip charts), indicating real time versus temperature, shall be utilized to document all times and temperatures above 450°F.

10.4.5.4 Intermediate stress relief. For weld grooves over 4 inches deep, intermediate stress relief may be used after one half of the weld groove depth has been filled.

10.4.6 Hardness testing for stress relief. When tempering temperature information is not available, hardness surveys shall be performed before and after stress relief. Hardness readings shall be taken from a portion of the base material to be subjected to a full stress relief cycle. A minimum of five hardness readings shall be taken both before and after stress relief from the same region of the shaft. Standard Rockwell (B or C scale) or Brinell (standard ball) hardness testing is acceptable. Microhardness (for example Diamond Pyramid) testing is not acceptable. Other hardness testing methods may be used with approval (see 14.4). Information generated by hardness testing requires approval (see 14.4).

10.4.7 Welding position. Unless otherwise approved (see 14.4), major and minor repairs shall be made in the flat position while rotating the shaft with the axis in the horizontal direction.

10.4.8 Distortion measurement during welding. When welding is expected to result in distortion in excess of the acceptable limit, distortion shall be measured during welding. Before preheat is applied, four dial indicators shall be located on the shaft. Two dial indicators, located 90 degrees apart, shall be placed equidistant on each side of the centerline of the weld joint. The dial indicators shall be located as far as practical from the area to be welded, but need not be more than 10 feet distant. Prior to application of preheat, the dial indicators shall be set to measure distortion in either direction. Appropriate means (for example, welding sequence, peening) shall be used to correct distortion during welding. After welding is completed and preheat is removed, distortion in the shaft shall be within limits specified on the applicable drawing.

11. STRAIGHTENING

11.1 Shaft material. This standard shall be followed for straightening of all steel shafts. Straightening of shafts not listed in 4.2.1 or 4.2.2 requires approval prior to straightening (see 14.4). This section is specifically applicable to shafting material listed in MIL-S-23284 (see 4.2.1 and 4.2.2). The method

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of straightening shall depend on available facilities, experience of operating personnel, processing condition of shafting, whether new or used shafting is involved, and the extent of straightening required.

11.1.1 New shafts. When new shafting requires straightening after the final heat treatment, the straightening process shall be followed by a stress relief performed at a minimum of 50°F below the final tempering temperature.

11.1.2 Used shafts. Except as specified in 11.1, used shafting may be straightened. The straightening process shall be followed by a stress relief performed at a minimum of 50°F below the final tempering temperature.

11.2 Procedure.

11.2.1 Determine location, magnitude, and type of bend. Prior to initiating straightening processes, the location, magnitude, and type of bend shall be determined. With the entire shaft supported on centers in a lathe, conduct a dial indicator survey of shaft run-out at stations spaced every few feet of shaft length to determine the magnitude and location of the bend. At each station, mark the position of maximum run-out on the periphery of the shaft. The marks on the shaft periphery will be in a straight line if the shaft contains a simple bend and will spiral part way around the shaft if the shaft is both bent and twisted. The maximum run-out shall be plotted on an exaggerated scale to locate the bend center and determine the extent of straightening required.

11.2.2 Methods.

11.2.2.1 Hot straightening. These methods are intended for use during manufacture of new shafts where the effects of scaling, local distortion due to pressing, and so forth, can be removed in subsequent machining. Other applications require approval (see 14.4).

11.2.2.1.1 During tempering. After the shaft has been held at the tempering temperature for the required time, the shaft shall be removed from the furnace and straightened under a press. The shaft shall be stress relieved after straightening (see 11.2.2.1.3).

11.2.2.1.2 After tempering. The shaft shall be charged into the furnace at 600°F or lower, raised to a temperature 50 to 100°F below the tempering temperature, equalized for 1 hour per inch of thickness, then removed and hot straightened under a press. The shaft shall be stress relieved after hot straightening (see 11.2.2.1.3).

11.2.2.1.3 Stress relief. After hot straightening, the shaft shall be immediately stress relieved at a temperature at least 50°F below the tempering temperature but not less than 1050°F for 2 hours per inch of maximum thickness, but in no case for less than 2 hours. Stress relief shall be followed by furnace cooling to 600°F under uniform conditions at a rate not greater than 500°F per hour divided by the maximum thickness of the part in inches, but in no case greater than 500°F per hour.

11.2.2.1.4 Cold rolling. Cold rolling of the shaft shall be performed after hot straightening and stress-relieving.

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11.2.2.2 Localized heating (hot spotting). The hot spotting method is intended for straightening used shafting only. Hot spotting involves the iterative use of alternate heating and cooling cycles and provides a means of straightening a section of shafting aboard ship with a minimum of machinery and equipment. Whereas this method offers distinct advantages, it shall be employed with caution so that the shaft is not cracked or damaged. Magnetic particle or dye penetrant testing, as applicable, as specified in 7.2 shall be performed after hot spotting is completed.

11.2.2.2.1 Hot spotting procedure. The following hot spotting procedure shall be used. Determine the location and magnitude of the bend in the shaft as specified in 11.2.1. Position the bent section of the shaft with the bend on top (knee-up). Locate and monitor dial indicators as specified in 11.2.2.2.2. Apply a concentrated local heat in a circular fashion over a diameter of about 2 to 3 inches at or near the top of the bend. As the metal expands with heat, the bend will increase. The expanding metal will be compressed by the restraining effects of the rest of the shaft. Compressive stresses will be increased with heat input until compressive yielding takes place. When cooled, tensile stresses will develop in the area in which yielding occurred. Tensile stresses will act to oppose the bend and partial straightening will occur. One application may be adequate for small bends while larger bends may require a series of successive spots spread over an area near the maximum run-out. The shaft shall be allowed to cool between each spot and the run-out shall be plotted as specified in 11.2.1. This allows evaluation of the effect of the previous hot spot and helps to locate where the next spot should be applied. Post repair stress relief shall not be performed after hot spotting. Cold rolling (see 12.1.2) may not be required.

11.2.2.2.2 Dial indicators. Before applying local heat, four dial indicators shall be located on the shaft. Two dial indicators shall be placed on the top of the shaft each equidistant on each side of the bent region. Another dial indicator shall be placed 90 degrees from each of the first two. The dial indicators shall be located as far as practical from the area to be hot spotted, but need not be more than 10 feet distant. Prior to application of heat, the dial indicators shall be set to measure distortion in either direction. After hot spotting is completed and the shaft has cooled to ambient temperature, total indicated run out in the shaft shall be a few thousandths of an inch, opposite to the original run-out.

11.2.2.3 Cold straightening. The methods specified in 11.2.2.3.1 through 11.2.2.4 are intended for use during final machining operations or where it is not feasible to hot straighten. They may be used on used shafting, and on new shafting with prior approval (see 14.4).

11.2.2.3.1 Cold peening. The location and magnitude of the bend in the shaft shall be determined as specified in 11.2.1. The shaft shall be supported approximately 1 to 2 feet on each side of the bend center and positioned so that the high spot of the bend is downward (at 6 o'clock). The shaft shall be peened on the top with sledge hammers, starting at a point over the center of the bend and continued over the shaft in both directions as far as the supports. Peening will raise the bent portion of the shaft to a common axis with the ends. After machining as specified in 11.2.2.3.1.1, additional straightening shall utilize protective copper or steel plates, of sufficient thickness to prevent marring of the shaft surface, placed on the shaft surface where blows are to be struck.

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11.2.2.3.1.1 Machining or grinding after cold straightening. When the shaft is within straightness requirements of the applicable shafting drawing, the shaft shall be machined or ground to remove any inadvertent peening marks, but in no case shall the shaft diameter be reduced by more than as specified in 10.3.1. If shafts are machined or ground in regions which require cold rolling in accordance with the applicable shafting drawing, shafts shall be cold rolled after machining. After machining, shafts shall be checked for straightness as specified in 11.2.1 and re-straightened if necessary.

11.2.2.3.2 Vibration support. The shaft should be supported for straightening by vibration as described below (see figure 3). Methods used to support the shaft need not be limited to those specified as follows:

(a) Fasten one end of the shaft to the face plate of a lathe or other device. Suspend the shaft at the center of bend from an overhead crane with the maximum run-out facing down. Allow the other end to hang free.
(b) Suspend the shaft from an overhead crane with the maximum run-out facing down and both ends hanging free.
(c) Support the shaft, with the run-out facing down, on blocks on either side of the center of bend and approximately 3 feet apart.
(d) Support the shaft in saddles on either side of the center of bend with the run-out facing down, and the point of maximum run-out supported with a jack.

11.2.2.3.2.1 Vibratory forces. After the shaft has been supported properly, it shall be vibrated (violently) by striking it on top at both ends with a heavy machinist's hammer or a light sledge. The combination of vibratory forces, and bending moments introduced by selected support points will cause the shaft to return to the original neutral axis. Protective plates shall be required as specified in 11.2.2.3.1. Machining or grinding shall be as specified in 11.2.2.3.1.1. If applicable, re-cold rolling of the shaft shall be performed after machining or grinding (see 11.2.2.3.1.1).

11.2.2.4 Other methods. For alloy steels (see 4.2.2), straightening methods other than as specified herein require approval (see 14.4). For carbon steels (see 4.2.1), methods require review and approval by the cognizant authorized representative (see 14.4) prior to straightening.

12. COLD ROLLING

12.1 General requirements. This section is applicable to shafting materials listed in 4.2.1 and 4.2.2 and shall be used as a guide for cold rolling of all steel shafting.

12.1.1 New shafts. When specified on applicable drawings, cold rolling of propulsion shafting shall be performed on the after end of new propulsion shaft sections. Cold rolling of new shaft sections shall be accomplished prior to cutting propeller keyways or installation of after strut sleeves.

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12.1.2 Refurbished shafts. For refurbished shafts, cold rolling shall be performed on previously cold rolled surfaces which have been welded, hot straightened, cold straightened and machined or ground, stress relieved, or otherwise thermally treated. If possible, cold rolling of refurbished sections shall be accomplished prior to installation of after strut sleeves. Dummy keys (see 12.2.5) shall be fit to tapers prior to cold rolling. Cold rolling need not be accomplished on shafting that is straightened aboard ship.

12.2 Procedure.

12.2.1 Finish requirements. The area of the propeller shaft to be cold rolled shall have a 125 microinch root mean square or finer finish before rolling.

12.2.2 Roller sizes. Cold rolling shall be accomplished using a 9-inch diameter hardening roller with a 1-1/2 inch crown radius bearing on the shaft. The hardening roller shall be opposed on the opposite side of the shaft by a 9-inch diameter smoothing roller with a 4-inch crown radius.

12.2.3 Rotation and feed. The shaft shall be rotated with a surface speed of about 60 feet per minute and shall be moved through the rollers at a rate of feed of about 1/16 inch for each revolution of the shaft.

12.2.4 Shafts.

12.2.4.1 New shafts. Cold rolling of new propeller shaft sections shall be continuous from a point 12 inches forward of the forward end of the after shaft sleeve to a point 2 inches forward of the after end of the taper. One pass shall be made over the area to be rolled.

12.2.4.2 Repaired shafts. For existing shaft sections where it is necessary to reroll the shaft sections (see 12.1.2) rolling shall end 12 inches on either side of the area affected by the operation listed in 12.1.2.

12.2.5 Rolling tapered sections of existing shaft. When rerolling of an existing tapered section is required, dummy keys shall be fitted to fill keyways. The spooned out area of the forward end of the keyway shall be filled with weld metal and stress relieved prior to rolling to provide a continuous surface for the rollers to bear upon. The welding procedure shall be as specified in section 5. The after end of the dummy key shall also be welded in place as necessary to prevent movement during rolling. The keyway sled runner end shall be remachined and faired to shaft diameter when the dummy key is removed after rolling.

12.2.6 Lubrication. The area to be cold rolled shall be coated with an extreme pressure lubricant during rolling.

12.2.7 Roller loads. Roller loads shall be as shown in table III.

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TABLE III. Roller loads. 1/

Yield point (lb/in^2) (0.01 percent offset)	Roller load (pounds)	Material
30,000	18,500	
35,000	20,000	MIL-S-23284, class 4
40,000	21,500	
45,000	26,500	MIL-S-23284, class 3
50,000	30,000	
55,000	32,750	MIL-S-23284, class 2
60,000	36,000	
65,000	40,000	
70,000	43,400	
75,000	46,000	MIL-S-23284, class 1, class 5
80,000	49,000	
85,000	52,000	
90,000	55,000	MIL-S-23284, class 6

1/ The roller load on the shaft shall be increased slowly from zero to the load specified in table III starting 2 inches ahead of the section to be rolled, and decreased gradually from the load specified in table III to zero over a distance of 2 inches following the section to be rolled.

12.2.8 Inspection. Upon completion of cold rolling and machining the keyways, the rolled area of the shaft shall be inspected by magnetic particle inspection as specified in 7.2.

12.2.9 Final finishing. The amount of metal removed for final finishing after cold rolling shall be kept to a minimum and shall not be greater than 0.005 inch on the diameter. The reduction of the radius due to cold rolling shall be 0.001 to 0.002 inch.

13. SHOT PEENING REQUIREMENTS

13.1 Shot peening requirements. Flange fillet area which has been weld repaired shall be shot peened after stress relief. Peening shall be in accordance with MIL-S-13165 or MIL-R-81841 and to the intensity specified therein. The peening shall include the entire curvature of the fillet and the entire circumference.

14. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

14.1 Intended use. This standard is intended for use in weld cladding, repair welding, mechanical straightening, cold rolling, workmanship, inspection, and record keeping for main propulsion shafting for ships of the United States Navy.

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14.1.1 This standard may be utilized as a guide for repair of shafts other than main propulsion shafting.

14.2 Issue of DODISS. When this standard is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation (see 2.1.1 and 2.2).

14.3 Facility approval. Facilities intending to repair shafts for the U.S. Navy as specified in this standard should have prior NAVSEA approval of their technical ability for compliance. This approval should be arranged prior to acceptance of work and may include documentation review and facility audits.

14.4 NAVSEA authorized representative. A NAVSEA authorized representative is any Government representative specifically authorized to approve equipment, material, or procedures within the scope of this standard for NAVSEA. These representatives are as follows:

(a) For Government shipyards: The delegated representative of the shipyard Commander.
(b) For commercial shipyards: The delegated representative of the Supervisor of Shipbuilding, Conversion and Repair (SUPSHIP), or the American Bureau of Shipping when specified in the ship's specifications for a particular ship. This includes all applicable areas in the shipyard and applicable items furnished to the shipyard by subcontractors.
(c) For Government purchase items: The delegated representative of the Commanding Officer, Naval Ship Systems Engineering Station (NAVSSES), Philadelphia.
(d) When delegated by (a), (b), or (c), the representative of the Defense Contract Administration Services Management Area (DCASMA).
(e) Technical representative specifically authorized by NAVSEA.

Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

14.5 Subject term (key word) listing.

Cladding
Gas metal arc
Heat-affected zone
Liquid penetrant inspection
Magnetic particle inspection
Nondestructive testing
Peening
Shielded metal arc
Ultrasonic inspection

Preparing activity:
Navy - SH
(Project 2010-N023)

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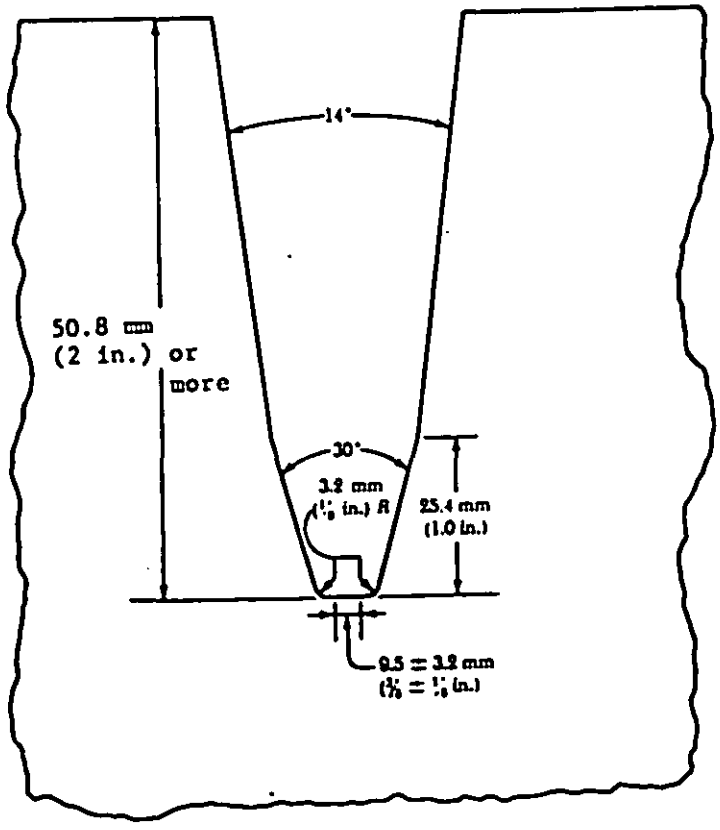


FIGURE 1. Joint design for weld repair in solid shafting.

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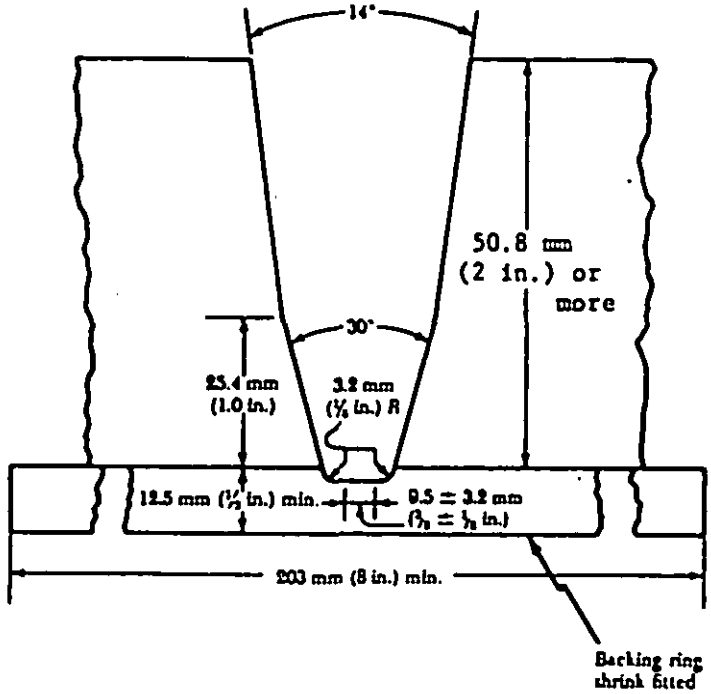


FIGURE 2. Joint design for weld repair in hollow shafting.

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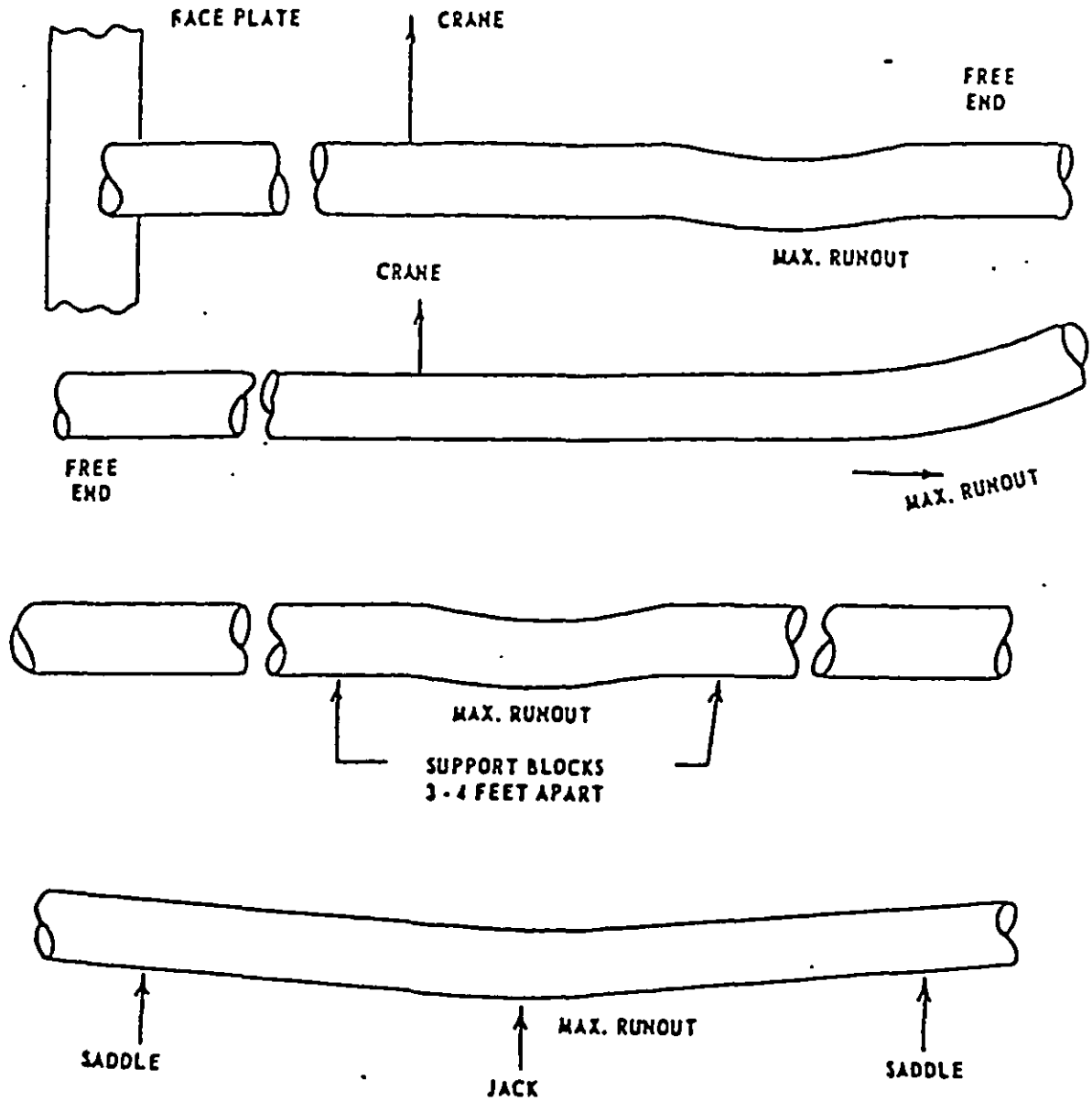


FIGURE 3. Methods of supporting shaft for straightening by vibration.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-2191(SH)	2. DOCUMENT DATE (YYMMDD) 1992 November 5

3. DOCUMENT TITLE Repair Welding, Weld Cladding, Straightening and Cold Rolling of Main Propulsion Shafting

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL POINT OF CONTACT (TPOC): Mr. Albert Grubowski, SEA 05M2 PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: (703) 602-0143	(2) AUTOVON 8-332-0143
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command ATTN: 05Q42, 2531 National Center, Bldg. 3 Washington, DC 20362-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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